

Multiple Choice

1. **Answer:** D

Options: A) $(1/2) k T$; B) kT ; C) $(3/2) k T$; D) $3 kT$

Note: Correct option: D - $3 kT$

2. **Answer:** A

Options: A) real; B) imaginary; C) degenerate; D) linear

Note: Correct option: A - real

3. **Answer:** B

Options: A) an average of 10 times, with an rms deviation of about 4; B) an average of 10 times, with an rms deviation of about 3; C) an average of 10 times, with an rms deviation of about 1; D) an average of 10 times, with an rms deviation of about 0.1

Note: Correct option: B - an average of 10 times, with an rms deviation of about 3

4. **Answer:** A

Options: A) decreased by a factor of 9; B) decreased by a factor of 49; C) decreased by a factor of 81; D) increased by a factor of 9

Note: Correct option: A - decreased by a factor of 9

5. **Answer:** D

Options: A) $0.1 \text{ GeV}/c^2$; B) $0.2 \text{ GeV}/c^2$; C) $0.5 \text{ GeV}/c^2$; D) $1.0 \text{ GeV}/c^2$

Note: Correct option: D - $1.0 \text{ GeV}/c^2$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $3(1/2)mc$ '.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Hall coefficient. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Constant volume. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key